PPI Edge Infused Spatial–Spectral Adaptive Residual Network for Multispectral Filter Array Image Demosaicing

Bowen Zhao, Jiesi Zheng, Yafei Dong, Ning Shen, Jiangxin Yang, Yanlong Cao, and Yanpeng Cao, *Senior Member, IEEE*

***Abstract*— Multispectral filter array (MSFA) sensors provide a cost-effective and one-shot acquisition solution to obtain well-aligned multiband images, which are helpful for various optical and remote-sensing applications. However, the sparse spatial sampling rate and strong spectral cross correlation make MSFA image demosaicing a challenging problem. Therefore, it is essential to develop effective MSFA demosaicing solutions to reconstruct full-resolution and high-fidelity multispectral images from the raw mosaic image. In this article, we present a pseudo-panchromatic image (PPI) edge-infused spatial–spectral adaptive residual network (PPIE-SSARN) for MSFA image demosaicing. The proposed two-branch model deploys a residual subbranch to adaptively compensate for the spatial and spectral differences of reconstructed multispectral images and a PPI edge infusion subbranch to enrich the edge-related information. Moreover, we design an effective mosaic initial feature extraction module with a spatial- and spectral-adaptive weight-sharing strategy whose kernel weights can change adaptively with spatial locations and spectral bands to avoid artifacts and aliasing problems. Experimental results demonstrate the superiority of our proposed method, outperforming the state-of-the-art MSFA demosaicing approaches and achieving satisfying demosaicing results in terms of spatial accuracy and spectral fidelity. Our models and code will be publicly available.**

***Index Terms*— Deep learning, image demosaicing, multispectral filter array (MSFA), multispectral imaging.**

I. Introduction

IT IS highly desirable to build multispectral imaging systems for obtaining complementary information on scenes or objects in different spectral bands to facilitate various optical and remote-sensing applications such as land-cover classification [1], [2], food quality analysis [3], crop monitoring [4], and medical diagnosis [5]. Most existing solutions deploy multiple image sensors or replaceable optical filters for multispectral image acquisition [6], [7]. However, such designs will not only incur more hardware costs, but is also time-consuming and not suitable for dynamic scene shooting.

In recent years, the multispectral filter array (MSFA) sensor has attracted increasing attention and becomes a promising approach to simultaneously capture high-quality and well-aligned multispectral images during one-shot acquisition [8], [9]. The use of the MSFA sensor has the potential to provide a lightweight, cost-effective, and simultaneous multispectral-exposure alternative to traditional push-broom acquisitions for land-cover classification and crop monitoring. These MSFA-based acquisition devices can be mounted on drones, airborne platforms, or mini-satellite constellations, making them particularly attractive for remote-sensing applications [10], [11], [12], [13].

Inspired by the Bayer color filter array (CFA) [14], an MSFA is formed with several bandpass optical filters arranged in a spatially repetitive pattern covered on the monochrome imaging sensor, for example, CMOS sensor, as illustrated in Fig. 1. The captured original image is called the raw mosaic image, in which an individual pixel only records sparse data of a particular band due to the spatial subsampling of all spectral bands in the MSFA. Consequently, the missing spectral values must be estimated from the acquired sparse spatial data to obtain full-resolution and high-fidelity multispectral images. This process is typically referred to as MSFA demosaicing [15], which provides an indispensable component for reconstructing multispectral images in high-quality MSFA single-sensor imaging systems [16].

Unlike the well-studied traditional CFA demosaicing problems [17], [18], the high-accuracy demosaicing of MSFA sensor-captured images represents a challenging task for two major reasons. First, increased arrayed spectral bands and the absence of a dominant band indicate a lower spatial sampling rate, which weakens the spatial correlation among the neighboring pixels sampled by the same band. Sparsely sampled raw data will decrease the spatial accuracy of reconstructed multispectral images. Second, since the spectral bands become narrower as the spectrum is further subdivided, the spectral cross correlation among nearby bands is more potent than the one among distant ones, which causes aliasing problems among different channels in the reconstructed multispectral images [19].

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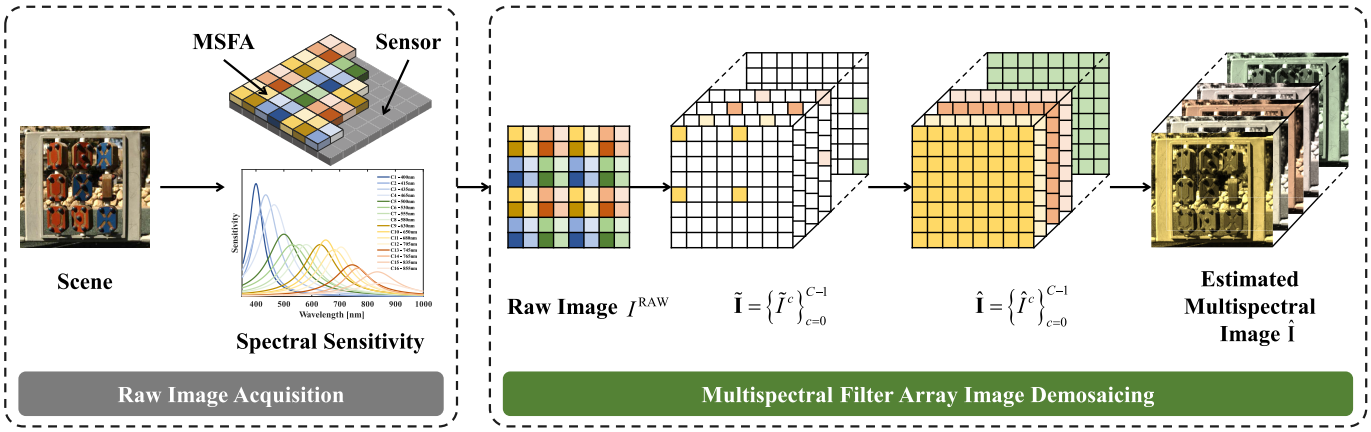


Fig. 1. Raw image acquisition process and demosaicing model of a single-sensor multispectral camera fitted with MSFA.

Previously, researchers have investigated various handcrafted approaches to MSFA image demosaicing [20], [21], [22], [23], [24], [25], [26], [27]. Brauers and Aach [15] extended the CFA demosaicing to MSFA and proposed the weighted bilinear (WB) interpolation method. Chini et al. [28] first presented the pseudo-panchromatic image (PPI), the average image of overall spectral channels. Taking the spatial and spectral correlation of MSFA image into account, Mihoubi et al. [29], [30] proposed the PPI difference (PPID) method to facilitate MSFA demosaicing by sharpening the PPI using a channel residual structure. Such design also drives us to model the MSFA image demosaicing by deploying a residual structure to compensate for spatial–spectral differences of reconstructed multispectral images.

Inspired by the recent success of deep convolutional neural networks (CNNs) in various MSFA-based image applications such as object tracking [31], [32], image denoising [33], and image super-resolution [34], [35], some researchers attempted to build data-driven approaches to achieve high-accuracy MSFA image demosaicing [36], [37], [38]. Shinoda et al. [39] presented a deep demosaicing model based on Residual Network (ResNet) and 3-D convolution. Feng et al. [40] proposed the end-to-end mosaic convolution-attention network (MCAN) for MSFA image demosaicing, which proved the great significance of initial feature extraction on the raw mosaic image. However, the mosaic convolution module (MCM) in MCAN is based on a spectral periodic weight-sharing strategy that ignores the variability of spatial locations, leading to checkerboard distortion. By combining the advantages of the traditional PPI-based method, Liu et al. [41] presented the Deep PPI Demosaicing Network (DPDNet), demonstrating that the high-frequency information in PPI and the reconstructed multispectral images are highly related. However, the DPDNet did not give a specific solution for accurately extracting the high-frequency information from PPI.

Based on the above literature analysis, we make the following observations: 1) the spatial–spectral residual compensation structure can facilitate the reconstruction of high-quality multispectral images, as proved in PPID [30]; 2) it is desirable to design a particular initial feature extraction module for MSFA raw mosaic images, which is sensitive to spatial and spectral variations to avoid artifacts and aliasing problems [40]; and 3) it is important to extract high-frequency information (e.g., edge-related information) from PPI and ensure that no other redundant features interfere in the demosaicing process [41].

In this article, we present a data-driven framework for effective MSFA demosaicing and demonstrate its effectiveness for high-performance multispectral image reconstruction. First, we propose a two-branch CNN model for MSFA image demosaicing, which facilitates MSFA demosaicing using a novel spatial–spectral adaptive residual architecture. We also present a prior-based technique to compensate edge-related information under the guidance of PPI to help recover edge details of the reconstructed images. Moreover, we design an effective mosaic initial feature extraction module with a spatial- and spectral-adaptive weight-sharing strategy whose kernel weights adaptively change with spatial locations and spectral bands when sliding to avoid artifacts and aliasing problems. As a result, the proposed methodology can achieve high-accuracy MSFA demosaicing and reconstruct multispectral images in high quality with satisfactory spatial accuracy and spectral fidelity. The contributions of this article are summarized as follows.

1) We present a novel end-to-end CNN model for MSFA image demosaicing. The proposed two-branch architecture deploys a residual subbranch to adaptively compensate for the spatial and spectral differences of reconstructed multispectral images and a PPI edge infusion subbranch to enrich the edge-related information, facilitating high-accuracy MSFA demosaicing and high-quality multispectral image reconstruction.
2) We propose an effective mosaic initial feature extraction method, named decoupled dynamic mosaic filter (DDMF), with a spatial- and spectral-adaptive weight-sharing strategy. The convolution kernel weights generated by the DDMF module change adaptively with spatial locations and spectral bands when sliding. The undesired artifacts and aliasing problems can be avoided by decoupling the spatial and spectral information during the MSFA demosaicing process.
3) We set up extensive experiments to verify the effectiveness of the proposed methodology.

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Compared with other state-of-the-art MSFA demosaicing approaches [15], [30], [39], [40], [41], our proposed PPI edge infused spatial–spectral adaptive residual network (PPIE-SSARN) model exhibits good generalization properties and achieves significantly improved demosaicing results in terms of spatial accuracy and spectral fidelity.

II. System Model and Preliminaries

A. MSFA Camera Demosaicing Model

A single-sensor multispectral camera equipped with MSFA provides a raw mosaic image I^{RAW} with a spatial resolution of $H \times W$ pixels. A single spectral band $c \in \{1, 2, \ldots, C\}$ is related to each pixel $p(i, j)$, $i \in \{1, 2, \ldots, H\}$, $j \in \{1, 2, \ldots, W\}$, according to the MSFA design. We define a multispectral image with C channels and full resolution as $\mathbf{I} = \{I^c\}_{c=1}^{C}$ (not available in practice). Then, the mosaic image I^{RAW} can been viewed as the result of $\mathbf{I}$ sampled by spectrum according to the pixel coordinates (i, j) as

$$I_{i,j}^{\mathrm{RAW}} = I_{i,j}^{c} \quad \text{s.t. } c = \sqrt{C} \cdot ((i-1) \bmod \sqrt{C}) + (j-1) \bmod \sqrt{C} + 1. \tag{1}$$

Each pixel of the raw mosaic image I^{RAW} only contains information of a specific spectral band, and values of $C-1$ other channels are unavailable.

By splitting the raw mosaic image I^{RAW} into C channels images $\tilde{\mathbf{I}} = \{\tilde{I}^c\}_{c=1}^{C}$ with full resolutions, values of a certain number of pixels in $\tilde{I}^c$ are available. Demosaicing is performed on each sparse image $\tilde{I}^c$ to estimate the value of blank pixels. A full-resolution and high-fidelity image $\hat{\mathbf{I}}$ with C channels is predicted at each pixel (i, j) as

$$\hat{\mathbf{I}}_{i,j} = \left(\hat{I}_{i,j}^{1}, \ldots, \hat{I}_{i,j}^{c-1}, \tilde{I}_{i,j}^{c}, \hat{I}_{i,j}^{c+1}, \ldots, \hat{I}_{i,j}^{C}\right) \tag{2}$$

where c is calculated based on (1).

Fig. 1 shows the raw image acquisition process and demosaicing model of a 4×4 MSFA camera ($C = 16$) with no dominant band. After receiving light reflected from a scene, each sensor pixel in MSFA responds in a specific band according to the spectral sensitivity curve. The raw image demosaiced in one-shot acquisition can obtain the estimated 16 channels full-resolution multispectral image $\hat{\mathbf{I}}$.

B. Weighted Bilinear Interpolation

The WB interpolation is widely recognized as the most conventional method for MSFA image demosaicing [15]. Different weights are assigned according to the distance from the central pixel. Using a well-designed low-pass filter, the estimated multispectral image $\hat{\mathbf{I}}_{\mathrm{WB}} = \{\hat{I}_{\mathrm{WB}}^{c}\}_{c=1}^{C}$ is generated by interpolating the sparse image $\tilde{\mathbf{I}} = \{\tilde{I}^c\}_{c=1}^{C}$ as

$$\hat{\mathbf{I}}_{\mathrm{WB}} = \tilde{\mathbf{I}} \circledast W \tag{3}$$

where $\circledast$ denotes the convolution operator, and W is a 7×7 filter defined as

$$W = \frac{1}{4}\begin{bmatrix}1\\2\\3\\4\\3\\2\\1\end{bmatrix} \cdot \frac{1}{4}\begin{bmatrix}1\\2\\3\\4\\3\\2\\1\end{bmatrix}^{T} = \frac{1}{16}\begin{bmatrix}1&2&3&4&3&2&1\\2&4&6&8&6&4&2\\3&6&9&12&9&6&3\\4&8&12&16&12&8&4\\3&6&9&12&9&6&3\\2&4&6&8&6&4&2\\1&2&3&4&3&2&1\end{bmatrix}. \tag{4}$$

Since the WB interpolation is rapid and accessible, we opt for this method to generate a preliminary 4×4 MSFA image demosaicing result as a base for the following spatial–spectral adaptive residual compensation.

C. Pseudo-Panchromatic Image Estimation

The ground-truth (GT) PPI I^{PPI} refers to the mean image over all channels of a full-resolution multispectral image [28]

$$I^{\mathrm{PPI}} = \frac{1}{C}\sum_{c=1}^{C} I^{c}. \tag{5}$$

Since the full-resolution multispectral image $\mathbf{I}$ is unavailable in practice, we can only estimate a preliminary $\bar{I}^{\mathrm{PPI}}$ from the raw mosaic image I^{RAW}. Based on the assumption that the values between neighboring pixels are strongly correlated, a smoothing filter M is designed for 4×4 MSFA PPI estimating [29], [30]

$$M = \frac{1}{64}\begin{bmatrix}1&2&2&2&1\\2&4&4&4&2\\2&4&4&4&2\\2&4&4&4&2\\1&2&2&2&1\end{bmatrix}. \tag{6}$$

The estimated $\bar{I}^{\mathrm{PPI}}$ is then computed by

$$\bar{I}^{\mathrm{PPI}} = I^{\mathrm{RAW}} \circledast M. \tag{7}$$

It has been demonstrated that a strong linear relationship exists between the high-frequency information in PPI and the demosaiced multispectral images [41]. As a result, the estimated PPI is utilized to complement edge-related information for the reconstructed multispectral images in our proposed model.

III. Methodology

This section presents a two-branch CNN architecture for MSFA image demosaicing. We first introduce the overall network framework of the proposed PPIE-SSARN and then describe how the DDMF module works in detail.

A. Network Framework

An overview of the proposed PPIE-SSARN is depicted in Fig. 2. It is an end-to-end two-branch network containing spatial–spectral adaptive residual compensation and PPI edge infusion subbranches. The residual subbranch first

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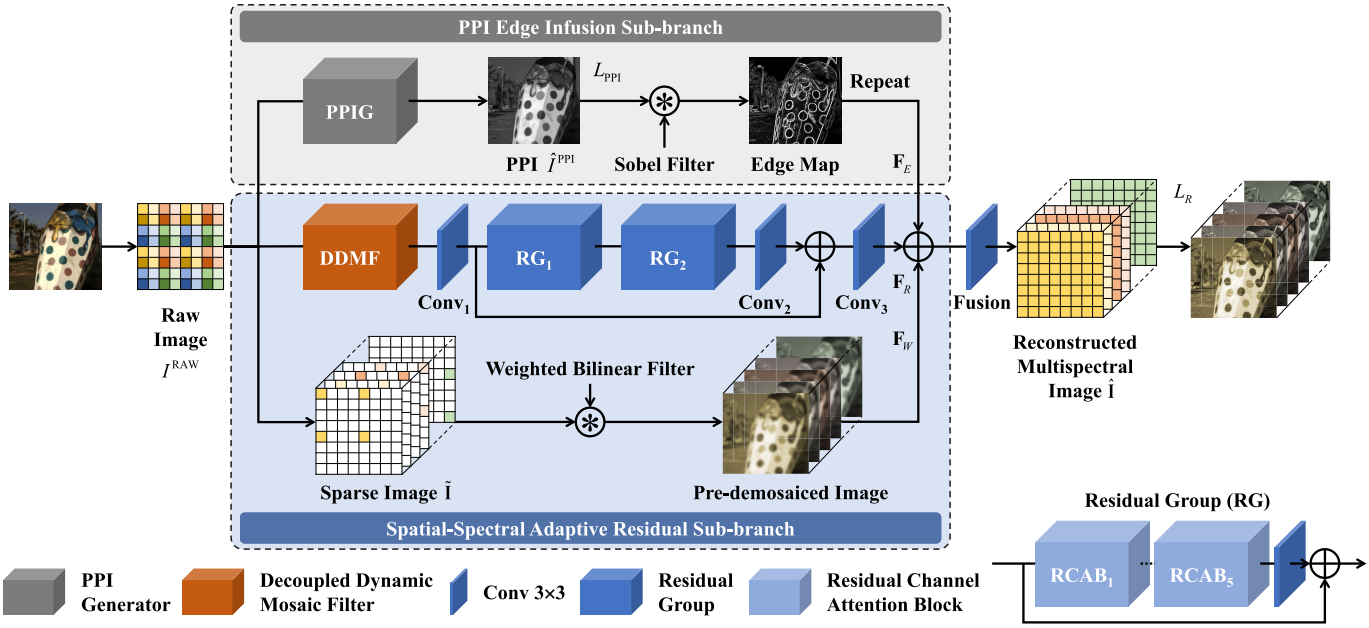


Fig. 2. Overall architecture of the proposed PPIE-SSARN. ⊕ means elementwise sum and ⊛ denotes the convolution operator.

makes use of WB interpolation to generate preliminary demosaiced images and then adaptively adjusts pixel values of the reconstructed multispectral images at different locations and channels by decoupling the spatial and spectral information. The convolution kernel weights generated by the DDMF module adapt to changes in spatial locations and spectral bands when sliding kernels. By adjusting the weights this way, the DDMF module can sensitively capture the difference information of various spatial locations and spectral bands, effectively compensating for the predemosaiced results. Moreover, the PPI edge infusion subbranch accurately estimates the PPI and integrates additional edge-related information to facilitate high-accuracy demosaicing of MSFA images in terms of spatial accuracy and spectral fidelity.

1) Spatial–Spectral Adaptive Residual Subbranch: Due to its rapidity and accessibility, we use the standard WB interpolation method to generate a preliminary 4 × 4 MSFA image demosaicing result as a base component for the following spatial–spectral adaptive residual compensation. Given the raw mosaic image I^{RAW} as input, the WB interpolation is performed on each sparse image $\tilde{\mathbf{I}} = \{\tilde{I}^c\}_{c=1}^{C}$ to generate the preliminary demosaiced multispectral image $\mathbf{F}_W = \{F_W^c\}_{c=1}^{C}$, as shown in Fig. 2 (bottom branch). This process can be expressed by the following equation:

$$F_W^c = \tilde{I}^c \circledast W \tag{8}$$

where W is the 6 × 6 WB filter defined in (7).

After obtaining the preliminary demosaiced image $\mathbf{F}_W$ via WB interpolation, it is essential to further complement the respective scarce knowledge based on the spatial and spectral variations of MSFA. For this purpose, we first integrate an effective mosaic initial feature extraction module (DDMF) in the spatial–spectral adaptive residual subbranch as shown in Fig. 2, providing a spatial- and spectral-adaptive initial feature extraction strategy to avoid the spatial artifacts and spectral aliasing problems. More details about the DDMF module are provided in Section III-B. Then, we adopt the residual group (RG) architecture [42] to extract deeper features for the spatial and spectral residual compensation. We stack two RGs which contain ten residual channel attention blocks (RCABs) (detailed architecture is illustrated in Fig. 3), and add long skip connections to stabilize the training of the deep network. The output features $\mathbf{F}_R = \{F_R^c\}_{c=1}^{C}$ are computed as

$$\begin{aligned}\mathbf{F}_D &= \text{Conv}_{3\times3}\left(H_{\text{DDMF}}\left(I^{\text{RAW}}\right)\right)\\ \mathbf{F}_R &= \text{Conv}_{3\times3}(\mathbf{F}_D + \text{Conv}_{3\times3}(H_{\text{RGs}}(\mathbf{F}_D)))\end{aligned} \tag{9}$$

where $\mathbf{F}_D$ is the features extracted by DDMF module and a 3 × 3 convolution layer. $H_{\text{DDMF}}(\cdot)$ denotes the function of DDMF module and $H_{\text{RGs}}(\cdot)$ denotes the function of two RGs. The detailed configurations of individual layers/modules in the spatial–spectral adaptive residual subbranch are shown in Table I.

2) PPI Edge Infusion Subbranch: Since the high-frequency information encoded in PPI and reconstructed multispectral image exhibits a strong positive linear relationship [41],

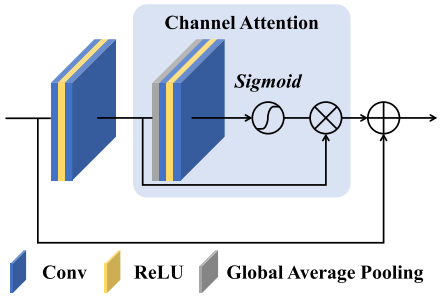


Fig. 3. Illustration of RCAB. ⊕ means elementwise sum and ⊗ denotes elementwise product.

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TABLE I

DETAILED CONFIGURATIONS OF INDIVIDUAL LAYERS/MODULES IN THE SPATIAL–SPECTRAL ADAPTIVE RESIDUAL SUBBRANCH. THE DIMENSIONS OF OUTPUT MAPS ARE INDICATED AS $C \times H \times W$, WHERE C IS THE CHANNEL NUMBER, H IS THE IMAGE HEIGHT, AND W IS THE IMAGE WIDTH

Modules			Output Size
Input (I^{RAW})			$1 \times 480 \times 512$
DDMF	DDMF, 5×5, 16		$16 \times 480 \times 512$
Conv_1	Conv, 3×3, 64		$64 \times 480 \times 512$
RG_1	RCAB_{1-5}	$\begin{bmatrix} \text{Conv}, 3 \times 3, 64 \\ \text{Conv}, 3 \times 3, 64 \\ \text{Channel Attention} \end{bmatrix} \times 5$	$64 \times 480 \times 512$
	Conv, 3×3, 64		$64 \times 480 \times 512$
RG_2	RCAB_{1-5}	$\begin{bmatrix} \text{Conv}, 3 \times 3, 64 \\ \text{Conv}, 3 \times 3, 64 \\ \text{Channel Attention} \end{bmatrix} \times 5$	$64 \times 480 \times 512$
	Conv, 3×3, 64		$64 \times 480 \times 512$
Conv_2	Conv, 3×3, 64		$64 \times 480 \times 512$
Conv_3	Conv, 3×3, 16		$16 \times 480 \times 512$

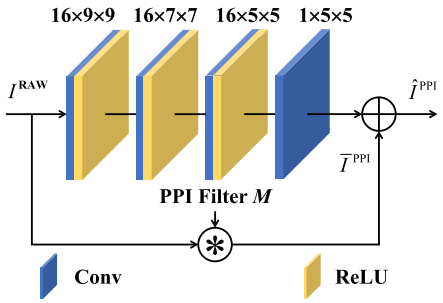


Fig. 4. Illustration of the PPIG module. $\oplus$ means elementwise sum and $\circledast$ denotes the convolution operator. The filter parameters are indicated as $C \times H \times W$, where C is the channel number, H is the kernel height, and W is the kernel width.

we propose to utilize the PPI for compensating the edge-related information of the demosaiced multispectral image. More specifically, we deploy a PPI edge infusion subbranch to extract high-frequency information from PPI and ensure that no other redundant features interfere with the demosaicing process.

As depicted in Fig. 4, we design a residual learning module called PPI generator (PPIG) by combining the CNN architecture with a traditional PPI estimation method mentioned in Section II-C. A preliminary $\bar{I}^{\text{PPI}}$ is generated by feeding the raw mosaic image I^{RAW} to the PPI filter M (defined in Section II-C). Then, four convolutional layers with different kernel sizes ($16 \times 9 \times 9$, $16 \times 7 \times 7$, $16 \times 5 \times 5$, and $1 \times 5 \times 5$) are deployed to estimate $\hat{I}^{\text{PPI}}$ by learning the residual between the preliminary $\bar{I}^{\text{PPI}}$ and the GT I^{PPI} as

$$L_{\text{PPI}}(\theta_p) = \frac{1}{HW} \sum_{i=1}^{H} \sum_{j=1}^{W} \left\| \hat{I}_{i,j}^{\text{PPI}} - I_{i,j}^{\text{PPI}} \right\|_1 \tag{10}$$

where θ_p is the parameter set of the PPIG module. H and W denote the height and width of the image, respectively. i and j are the pixel coordinates.

After obtaining the estimated $\hat{I}^{\text{PPI}}$, we adopt the Sobel filters [43] to perform gradient extraction and ensure that the high-frequency information of PPI is eventually utilized. As shown in Fig. 2 (top branch), the features $\mathbf{F}_E = \{F_E^c\}_{c=1}^{C}$ with edge-related information are calculated by

$$F_E^c = \hat{I}^{\text{PPI}} \circledast G_x + \hat{I}^{\text{PPI}} \circledast G_y \tag{11}$$

where $\circledast$ is the convolution operator. G_x and G_y are the Sobel filters in the x- and y-directions, respectively, which are defined as

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix}, \quad G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}. \tag{12}$$

The preliminary demosaicing result is obtained using $\mathbf{F}_W$, while the spatial- and spectral-adaptive residual information is extracted through deep feature extraction and represented by $\mathbf{F}_R$. Moreover, $\mathbf{F}_E$ provides additional edge-related information to enhance the sharpness and enrich the image details. After obtaining $\mathbf{F}_W$, $\mathbf{F}_R$, and $\mathbf{F}_E$, we deploy a 3×3 convolution layer to fuse all features and adjust the weights of different components to generate the reconstructed multispectral image $\hat{\mathbf{I}}$ as

$$\hat{\mathbf{I}} = \text{Conv}_{3\times3}(\mathbf{F}_W + \mathbf{F}_R + \mathbf{F}_E). \tag{13}$$

To make the training process easier to converge [44], we choose L_1 as our loss function. Given the full-resolution multispectral GT image $\mathbf{I}^{\text{GT}}$, the goal is to minimize the difference between $\hat{\mathbf{I}}$ and $\mathbf{I}^{\text{GT}}$ as

$$L_R(\theta) = \frac{1}{CHW} \sum_{c=1}^{C} \sum_{i=1}^{H} \sum_{j=1}^{W} \left\| \hat{\mathbf{I}}_{c,i,j} - \mathbf{I}_{c,i,j}^{\text{GT}} \right\|_1 \tag{14}$$

where θ is the parameter set of our proposed PPIE-SSARN. C, H, and W denote the channel numbers, image height, and image width, respectively. c, i, and j are the channel index and pixel coordinates, respectively.

Accordingly, we design a joint loss function denoted as L, which aims to minimize both the signal reconstruction errors of the PPI and the multispectral image simultaneously

$$L = L_{\text{PPI}} + L_R. \tag{15}$$

B. Decoupled Dynamic Mosaic Filter

Effective extraction of preliminary features from the raw mosaic image has been proven to be an essential aspect [40]. Since the standard convolution with global shared weights does not apply to the MSFA image, we propose the DDMF module with a spatial- and spectral-adaptive weight-sharing strategy following the design approach of [45]. By dynamically adjusting the convolution kernel weights based on the spatial and spectral characteristics, our proposed approach aims to effectively capture diverse and informative features from the input raw data. This adaptive mechanism helps

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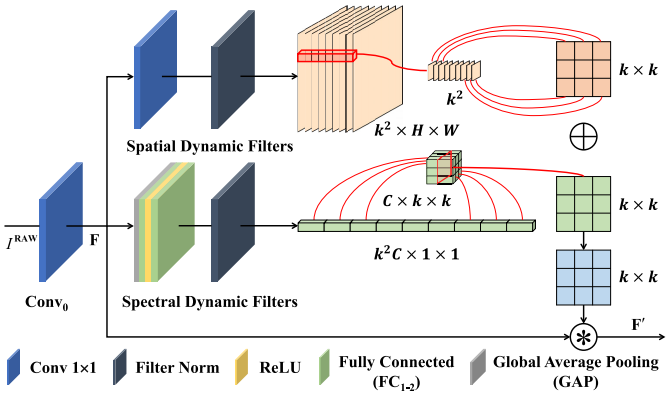


Fig. 5. Illustration of our proposed DDMF module. ⊕ means elementwise sum and ⊛ denotes the convolution operator.

mitigate spatial artifacts and spectral aliasing issues commonly encountered in traditional approaches. Notably, no downsampling operation is involved in this process, so the spatial information is well preserved.

Fig. 5 shows the illustration of the DDMF module. The raw mosaic image I^{RAW} is first fed to a 1×1 convolution layer for increasing dimension. Two separate attention-based branches are adopted for each pixel of a given spatial location and channel to generate spatial and spectral dynamic filters individually. These are then summed to form the convolution kernels and applied to the input features $\mathbf{F}$ at the relevant pixel position.

The spatial dynamic filters are predicted by a 1×1 convolution layer. According to the squeeze-and-excitation structure proposed in [46], the spectral branch consists of one global average pooling layer, one rectified linear units (ReLU) layer, and two fully connected layers. Filter normalization [45] is used to improve the stability of the training process in both branches. Given the input features $\mathbf{F} = \{F\}_{c=1}^{C}$, the output features $\mathbf{F}' = \{F'\}_{c=1}^{C}$ and the operation of the DDMF module can be defined as

$$F'_{u,c} = \sum_{v \in \Omega(u)} \left(D_u^{\mathrm{spa}}[p_u - p_v] + D_c^{\mathrm{spe}}[p_u - p_v] \right) \cdot F_{v,c} \qquad (16)$$

where $F'_{u,c} \in \mathbb{R}$ denotes the output feature value at the uth pixel and c^{th} channel, $\Omega(u)$ is the $k \times k$ convolution kernel around the uth pixel (we set $k = 5$ in our method so that the filter could cover the 4×4 MSFA pattern), and $F_{v,c} \in \mathbb{R}$ means the input feature value at the vth pixel and cth channel. $D[p_u - p_v]$ is the filter at position offset between the uth and v^{th} pixels: $[p_u - p_v] \in \{(-(k-1)/2, -(k-1)/2), (-(k-1)/2, -(k-1)/2 + 1), \ldots, ((k-1)/2, (k-1)/2)\}$, where p denotes the pixel coordinates. $D^{\mathrm{spa}} \in \mathbb{R}^{k^2 \times H \times W}$ denotes the spatial dynamic filter with $D_u^{\mathrm{spa}} \in \mathbb{R}^{k \times k}$ meaning the filter at u^{th} pixel. $D^{\mathrm{spe}} \in \mathbb{R}^{C \times k \times k}$ denotes the spectral dynamic filter with $D_c^{\mathrm{spe}} \in \mathbb{R}^{k \times k}$ meaning the filter at c^{th} channel.

The detailed configurations of individual layers/blocks and dimensional transformation during filter generation in the DDMF module are shown in Table II. Comparative results of different mosaic initial feature extraction methods are provided in Section IV-C.

TABLE II

DETAILED CONFIGURATIONS OF INDIVIDUAL LAYERS/BLOCKS AND DIMENSIONAL TRANSFORMATION DURING FILTER GENERATION IN THE DDMF MODULE. THE DIMENSIONS OF OUTPUT MAPS ARE INDICATED AS $C \times H \times W$, WHERE C IS THE CHANNEL NUMBER, H IS THE KERNEL HEIGHT, AND W IS THE KERNEL WIDTH

Modules		Output Size
Input (I^{RAW})		$1 \times 480 \times 512$
Conv$_0$	Conv, $1 \times 1, 16$	$16 \times 480 \times 512$
Spatial Dynamic Filters	Conv, $1 \times 1, 25$	$25 \times 480 \times 512$
	Reshape	$245760 \times 5 \times 5$ $HW \times k \times k$
Spectral Dynamic Filters	$\begin{bmatrix} \mathrm{GAP}, 16 \\ \mathrm{FC}_1, 4 \\ \mathrm{FC}_2, 400 \end{bmatrix}$	$400 \times 1 \times 1$
	Reshape	$16 \times 5 \times 5$ $C \times k \times k$

IV. EXPERIMENTS

A. Experimental Setup

1) Datasets: We evaluate our PPIE-SSARN model on two hyperspectral image datasets including one natural image dataset, that is, the ARAD-1K dataset [47], and one remote-sensing image dataset, the Chikusei dataset [48], which are listed as follows.

The NTIRE 2022 Spectral Demosaicing Challenge presents the ARAD-1K dataset. Compared with CAVE [49], TT59 [50], and TokyoTech [16], ARAD-1K is the first-of-its-kind large-scale dataset for MSFA demosaicing of natural scenes, containing 1000 images with 16 spectral bands covering 400–1000-nm wavelengths. The ARAD-1K dataset provides 16-channel full-resolution multispectral images with 480×512 spatial resolution as GT, which are spatially subsampled to generate the raw mosaic images using a 4×4 MSFA pattern with no dominant band. The ARAD-1K dataset is divided into 900 training images, 50 validation images, and 50 confidential test images (without corresponding GT). To evaluate our model quantitatively, we choose the 950 images with GT for training (900 images) and testing (50 images).

The Chikusei dataset is taken by Headwall Hyperspec-VNIR-C imaging sensor over agricultural and urban scenes in Chikusei, Ibaraki, Japan. Unlike the above datasets, it only contains one image consisting of 2517×2335 pixels and 128 bands in the spectral range from 363 to 1018 nm. In our experiments, we randomly crop 100 image patches with a resolution of 500×500 pixels and spatially subsample them to generate the raw mosaic images using the same 4×4 MSFA pattern as the ARAD-1K dataset. The Chikusei dataset is divided into 90 training images and ten test images.

2) Metrics: We adopt the peak signal-to-noise ratio (PSNR), structural similarity (SSIM) [51], and spectral angle mapper (SAM) [52] as the quantitative evaluation metrics. We utilize the PSNR to evaluate the pixel-level difference

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between the demosaiced multispectral image and the GT. SSIM assesses the image degradation between the demosaiced multispectral image and the GT. We also adopt the SAM to evaluate the spectral similarity between the demosaiced multispectral image and the GT. PSNR and SSIM increase with image quality, whereas SAM decreases with image quality.

3) Implementation Details: We implement the PPIE-SSARN model based on the PyTorch framework and train our model on NVIDIA RTX 2080Ti with CUDA 10.1 and cuDNN 7.6.5 for 4000 epochs. The Adam optimizer is utilized to optimize the weights by setting $\beta_1 = 0.9$, $\beta_2 = 0.999$, and $\epsilon = 10^{-8}$. Data augmentation is performed on the training images in both the ARAD-1K and Chikusei datasets, randomly cropped to 128×128 patches, rotated by 90°, 180°, 270°, and flipped horizontally or vertically. The batch size is set to 32. The initial learning rate is set to 10^{-4} and decreases to half every 500 epochs. The source code of the PPIE-SSARN model will be made publicly available in the future.

B. Comparisons With State-of-the-Art

We compare our proposed PPIE-SSARN model with a number of existing MSFA image demosaicing solutions quantitatively and qualitatively. We first consider two conventional methods including WB interpolation [15] and PPID [30]. We also consider three deep-learning-based models (3D-ResNet [39], DPDNet [41], and MCAN [40]), which are specially designed for MSFA image demosaicing. The source codes of these hand-crafted and CNN-based methods are either publicly available or reimplemented according to the original papers and retrained using the ARAD-1K and Chikusei datasets. Tables III–VI summarize the quantitative results and Figs. 6–10 show some qualitative comparisons.

1) ARAD-1K Dataset: Tables III–V show the quantitative results of PSNR, SSIM, and SAM assessments across all test images on the ARAD-1K dataset, respectively. For each metric, the best results are highlighted in bold. The second row of each table illustrates the average value of test images across all 16 bands, and the other rows show the values for each spectrum separately. It is indicated that our proposed PPIE-SSARN model outperforms the existing methods in terms of both image quality and spectral fidelity.

Fig. 6 shows visual comparisons of all methods to evaluate their demosaicing results subjectively. Some selected regions are magnified and highlighted in red boxes. It is observed that the conventional WB method performs unsatisfactorily in most cases as the estimated images are over-smoothed and blurred. On the contrary, PPID can achieve more accurate demosaicing in homogeneous regions but over-sharpens the edges and causes some artifacts, as shown in the highlighted regions of the second and fourth rows. Deep-learning-based approaches perform better overall. Horizontal and vertical stripes can be observed in the flat areas (the third row) of 3D-ResNet, while artifacts exist at the edges (the first and fourth rows). DPDNet can achieve better demosaicing results, but there are still some undesired streak artifacts at the edges (the first and fourth rows). The results of MCAN have some checkerboard distortion caused by the periodic feature extraction operation, as shown in the magnified regions of the third and fourth rows. In comparison, our proposed PPIE-SSARN model can achieve high-accuracy MSFA demosaicing and produce full-resolution multispectral images with better visual quality, compensating spatial–spectral residual information adaptively and recovering the edge details under the guidance of PPI.

To further demonstrate the superiority of our proposed method in recovering spatial and spectral information, the error maps between the GT and the demosaicing results of all approaches are visualized in Figs. 7 and 8. Two selected regions are highlighted in color boxes and zoomed in for each image. We show the visual comparisons for demosaicing results of three representative scenes at the same wavelength (415 nm) in Fig. 7. By comparing the reconstructed images (e.g., the "foot" pattern in the first row), our proposed method

TABLE III

PSNR ASSESSMENTS OF THE DEMOSAICED MULTISPECTRAL IMAGES AT DIFFERENT SPECTRAL BANDS ON THE ARAD-1K DATASET. NOTE THAT HIGHER PSNR VALUES INDICATE BETTER DEMOSAICING RESULTS. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**

	WB	PPID	3D-ResNet	DPDNet	MCAN	Ours
Average	36.582	41.279	41.642	44.818	46.466	**48.461**
400 nm	43.700	46.736	47.293	48.577	50.279	**52.264**
415 nm	42.652	46.012	46.441	48.365	49.898	**52.064**
435 nm	36.164	40.324	40.889	43.889	45.533	**47.426**
465 nm	33.849	38.570	39.581	42.469	44.059	**46.130**
500 nm	34.712	40.032	40.406	44.109	45.675	**47.574**
530 nm	34.545	40.374	41.309	44.625	46.333	**48.311**
555 nm	34.822	40.685	40.043	44.634	46.512	**48.560**
580 nm	34.318	40.440	40.823	44.821	46.469	**48.531**
630 nm	33.571	39.593	40.518	44.174	45.797	**47.918**
650 nm	33.675	39.444	38.744	43.698	45.378	**47.455**
680 nm	33.970	39.450	40.226	43.875	45.472	**47.528**
705 nm	34.342	39.444	39.532	43.354	44.930	**46.907**
745 nm	36.321	40.699	41.284	44.241	45.898	**47.879**
765 nm	38.619	42.409	42.643	45.190	46.872	**48.729**
835 nm	38.064	41.066	41.594	43.848	45.553	**47.474**
855 nm	41.991	45.081	44.938	47.213	48.807	**50.633**

TABLE IV

SSIM ASSESSMENTS OF THE DEMOSAICED MULTISPECTRAL IMAGES AT DIFFERENT SPECTRAL BANDS ON THE ARAD-1K DATASET. NOTE THAT HIGHER SSIM VALUES INDICATE BETTER DEMOSAICING RESULTS. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**. NOTE THAT WE HAVE EXPANDED THE SIGNIFICANT FIGURES OF THE DATA TO SHOW DIFFERENCES IN VARIOUS DEMOSAICING RESULTS

	WB	PPID	3D-ResNet	DPDNet	MCAN	Ours
Average	0.9212	0.9712	0.9756	0.9869	0.9903	**0.9936**
400 nm	0.9688	0.9837	0.9884	0.9904	0.9929	**0.9955**
415 nm	0.9642	0.9824	0.9870	0.9902	0.9927	**0.9954**
435 nm	0.9260	0.9672	0.9753	0.9849	0.9890	**0.9926**
465 nm	0.9063	0.9620	0.9711	0.9834	0.9877	**0.9919**
500 nm	0.9119	0.9691	0.9729	0.9867	0.9902	**0.9933**
530 nm	0.9096	0.9712	0.9755	0.9881	0.9913	**0.9941**
555 nm	0.9111	0.9722	0.9716	0.9882	0.9914	**0.9943**
580 nm	0.9065	0.9719	0.9751	0.9888	0.9915	**0.9944**
630 nm	0.8987	0.9691	0.9736	0.9879	0.9910	**0.9941**
650 nm	0.8989	0.9682	0.9665	0.9870	0.9903	**0.9937**
680 nm	0.9000	0.9676	0.9740	0.9874	0.9903	**0.9936**
705 nm	0.9014	0.9682	0.9698	0.9858	0.9894	**0.9929**
745 nm	0.9169	0.9688	0.9751	0.9855	0.9893	**0.9927**
765 nm	0.9336	0.9730	0.9775	0.9860	0.9897	**0.9930**
835 nm	0.9295	0.9657	0.9737	0.9822	0.9869	**0.9910**
855 nm	0.9562	0.9794	0.9829	0.9883	0.9912	**0.9941**

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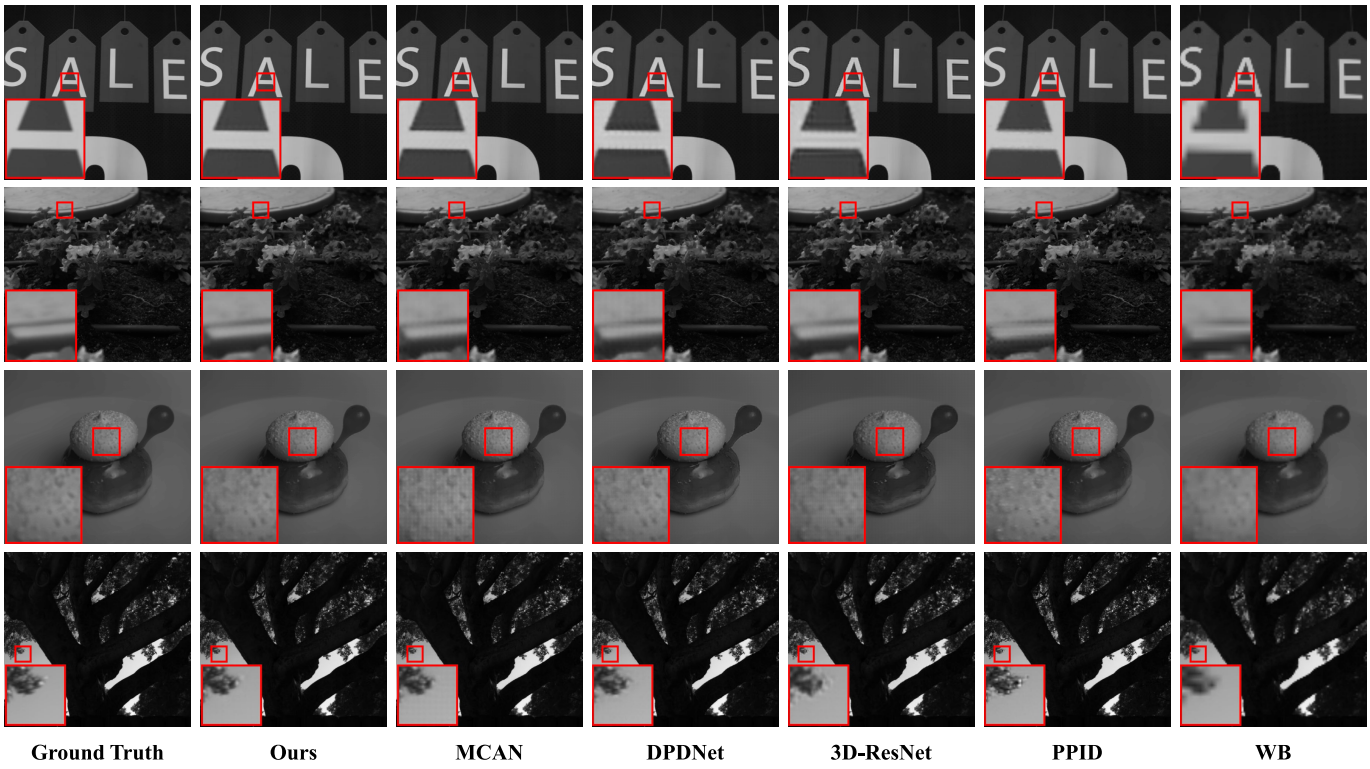


Fig. 6. Visual comparisons for demosaicing results on the ARAD-1K dataset. Our method is compared with five alternatives, that is, MCAN [40], DPDNet [41], 3D-ResNet [39], PPID [30], and WB [15]. The first column shows the GT. In addition, zoomed-in views of selected regions are provided. Please zoom in to see the details.

TABLE V

SAM ASSESSMENTS OF THE DEMOSAICED MULTISPECTRAL IMAGES AT DIFFERENT SPECTRAL BANDS ON THE ARAD-1K DATASET. NOTE THAT LOWER SAM VALUES INDICATE BETTER DEMOSAICING RESULTS. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**

	WB	PPID	3D-ResNet	DPDNet	MCAN	Ours
Average	12.274	7.289	6.476	4.768	3.970	**3.175**
400 nm	12.787	9.178	7.766	7.114	5.935	**4.743**
415 nm	13.009	8.964	7.792	6.650	5.583	**4.410**
435 nm	12.597	7.841	6.789	5.069	4.209	**3.406**
465 nm	12.607	7.305	6.290	4.547	3.837	**3.052**
500 nm	12.581	6.824	6.262	4.187	3.521	**2.838**
530 nm	12.490	6.360	5.529	3.838	3.176	**2.547**
555 nm	12.550	6.406	6.386	3.918	3.199	**2.554**
580 nm	12.537	6.190	5.464	3.611	3.000	**2.401**
630 nm	12.425	6.240	5.352	3.566	2.966	**2.350**
650 nm	12.453	6.414	6.563	3.816	3.186	**2.516**
680 nm	12.146	6.539	5.504	3.754	3.176	**2.521**
705 nm	12.016	6.414	6.066	4.156	3.477	**2.771**
745 nm	11.731	7.163	6.247	4.668	3.877	**3.103**
765 nm	11.724	7.724	6.880	5.413	4.495	**3.625**
835 nm	11.288	8.424	7.130	5.780	4.729	**3.823**
855 nm	11.442	8.451	7.606	6.205	5.150	**4.147**

can recover the complete details of edges and textures. In contrast, other methods produce obvious artifacts or blurs. It is proven that our proposed PPIE-SSARN model can accurately restore spatial information. Fig. 8 shows the visual comparisons for demosaicing results of a representative scene at various bands. Our proposed method generates the lowest absolute error between the GT and the demosaiced multispectral images in all bands. On the contrary, demosaicing results of other methods still need to improve on spectral aliasing problems, such as the reflection of the astronaut's helmet, which should not appear in the images at 705 and 745 nm. Our proposed PPIE-SSARN model can accurately extract the spatial–spectral differences and generate almost no aliasing problems, achieving higher spectral fidelity.

2) Chikusei Dataset: To comprehensively illustrate the effectiveness of our proposed PPIE-SSARN model, the hyperspectral remote-sensing Chikusei dataset is utilized for further evaluation. Table VI shows the average quantitative performance over ten testing images of all comparison approaches. We can observe that the proposed PPIE-SSARN model significantly outperforms other methods concerning all objective evaluation metrics. Similarly, the absolute error maps between the GT and the demosaiced multispectral images of all methods are visualized in Figs. 9 and 10. Whether comparing images of the same band in different scenes or images of different bands in the same scene, our proposed PPIE-SSARN model achieves the most satisfying demosaicing results in the better recovery of both edges and homogeneous regions. On the contrary, checkerboard distortion and artifacts can be observed in other comparison results.

Based on both quantitative and qualitative results shown in Tables III–VI and Figs. 6–10, it is experimentally verified that our proposed PPIE-SSARN model has the best performance in terms of objective and subjective evaluations.

C. Ablation Studies

In this section, we set up experiments to investigate the designs and effects of: 1) the two-stream network architecture;

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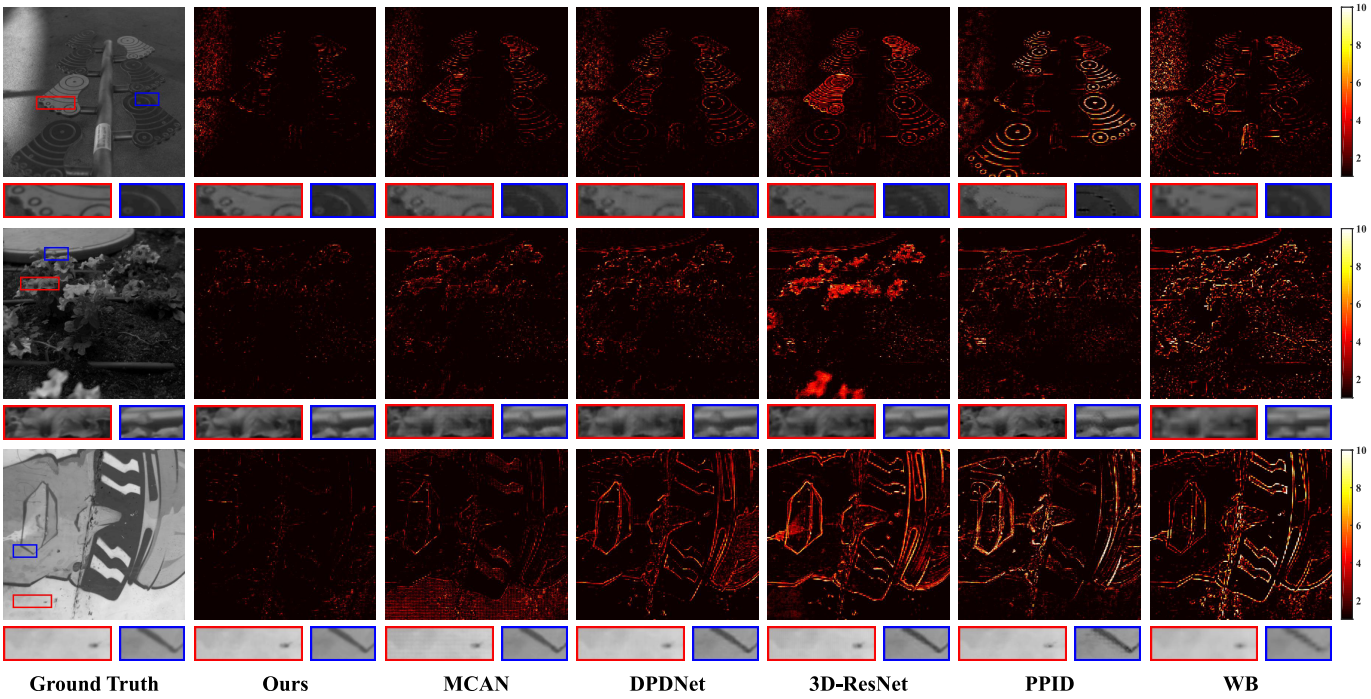


Fig. 7. Visual comparisons for demosaicing results of different scenes at one band (415 nm) on the ARAD-1K dataset. Our method is compared with five alternatives, that is, MCAN [40], DPDNet [41], 3D-ResNet [39], PPID [30], and WB [15]. The first column shows the GT. In addition, error maps and zoomed-in views of selected region images are provided. Please zoom in to see the details.

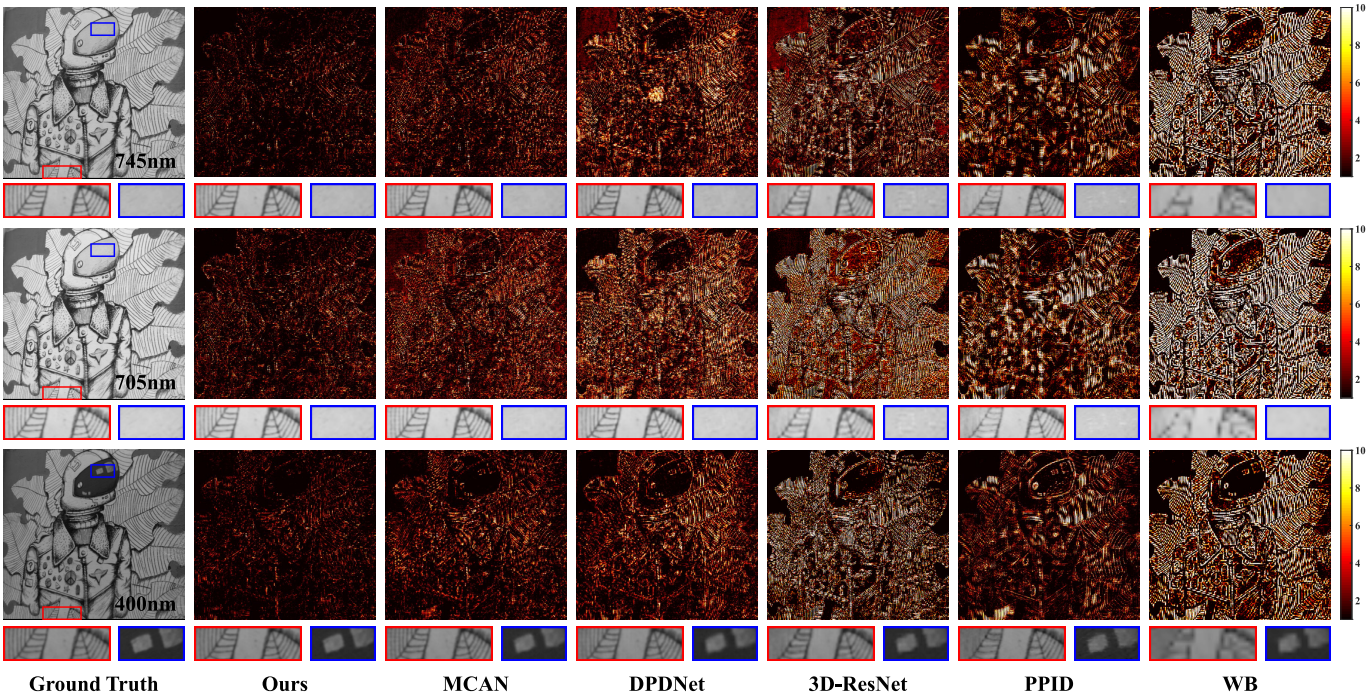


Fig. 8. Visual comparisons for demosaicing results of one representative scene at various bands on the ARAD-1K dataset. Our method is compared with five alternatives, that is, MCAN [40], DPDNet [41], 3D-ResNet [39], PPID [30], and WB [15]. The first column shows the GT. In addition, error maps and zoomed-in views of selected region images are provided. Please zoom in to see the details.

2) the initial feature extraction module; and 3) the PPI edge infusion strategy. The contributions of different components are analyzed via ablation experiments.

1) Two-Stream Network Architecture: The proposed model deploys a two-branch architecture consisting of a residual structure and a PPI edge infusion subbranch. The residual subbranch first makes use of WB interpolation to generate preliminary demosaicing results and then adaptively compensates the residual between the preliminary version and the GT. Moreover, the PPI edge infusion subbranch accurately estimates the PPI and integrates edge-related information to facilitate high-accuracy demosaicing of MSFA images.

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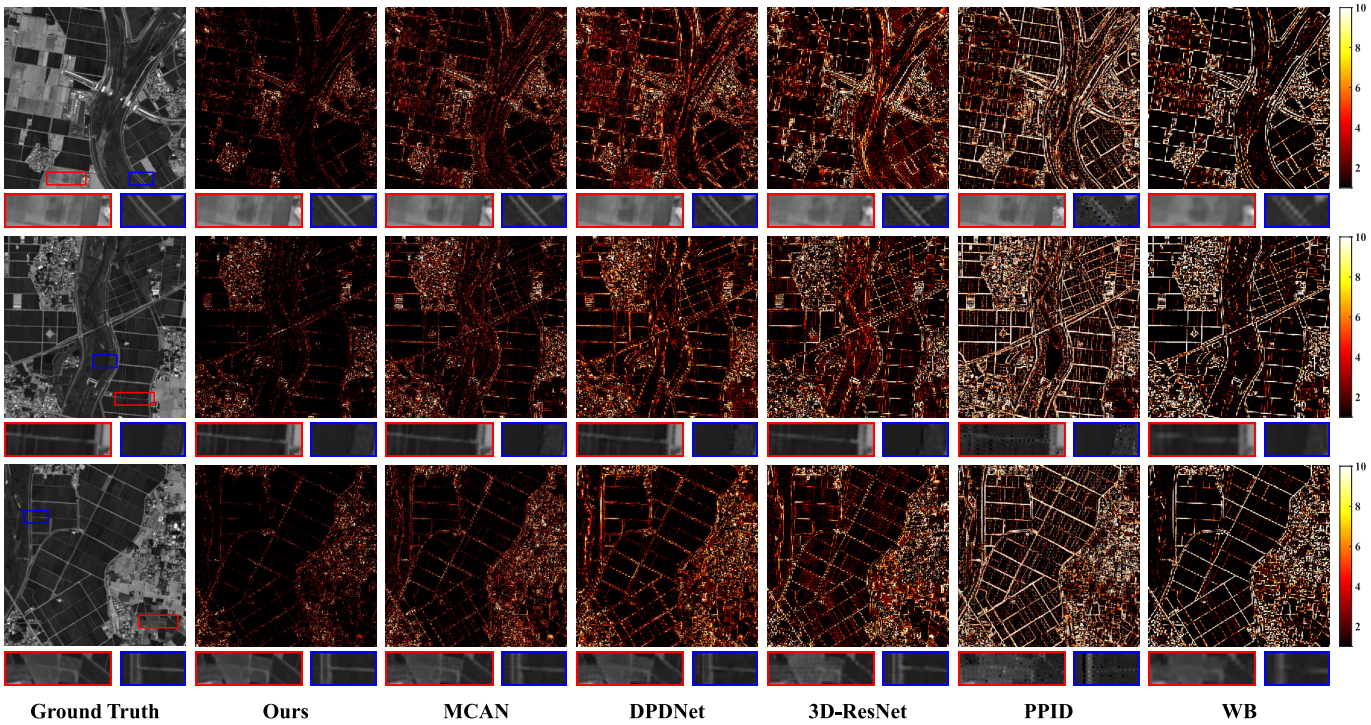


Fig. 9. Visual comparisons for demosaicing results of different scenes at one band (650 nm) on the Chikusei dataset. Our method is compared with five alternatives, that is, MCAN [40], DPDNet [41], 3D-ResNet [39], PPID [30], and WB [15]. The first column shows the GT. In addition, error maps and zoomed-in views of selected region images are provided. Please zoom in to see the details.

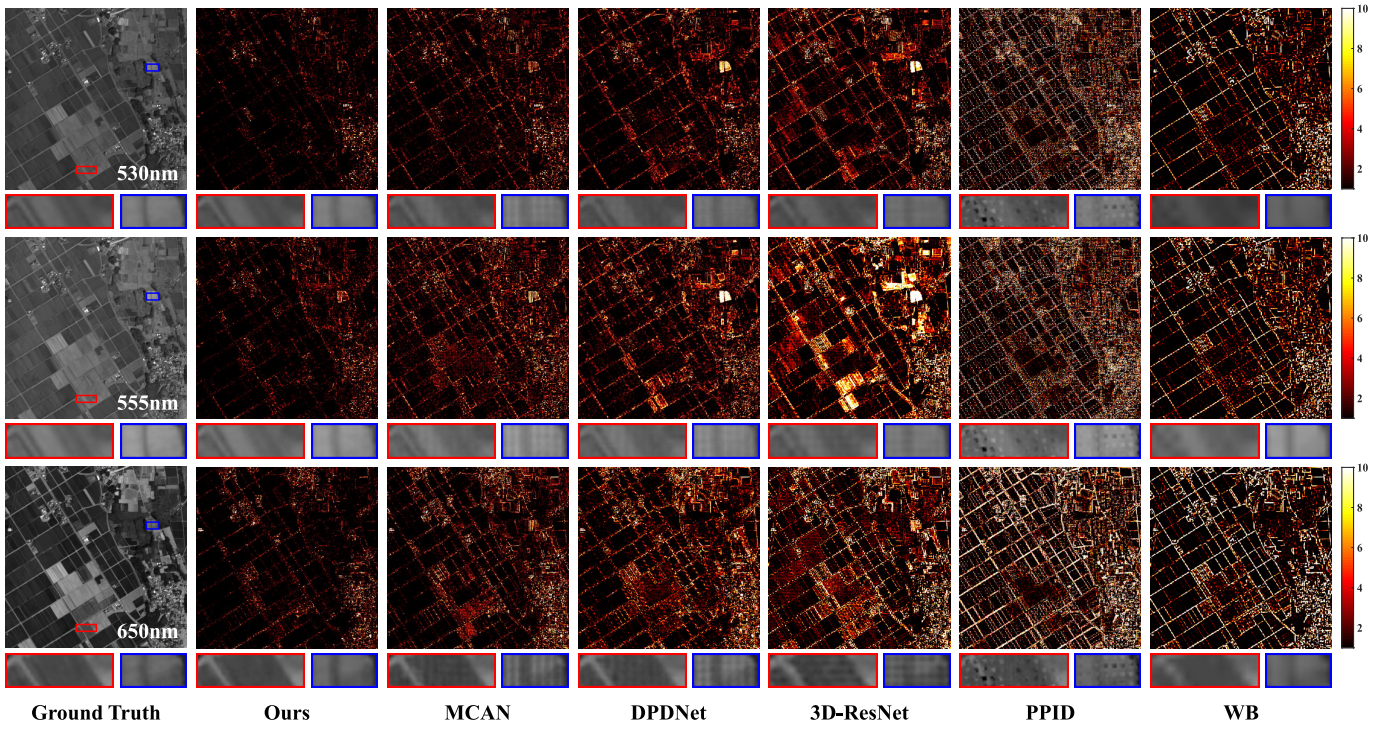


Fig. 10. Visual comparisons for demosaicing results of one representative scene at various bands on the Chikusei dataset. Our method is compared with five alternatives, that is, MCAN [40], DPDNet [41], 3D-ResNet [39], PPID [30], and WB [15]. The first column shows the GT. In addition, error maps and zoomed-in views of selected region images are provided. Please zoom in to see the details.

To demonstrate the effectiveness of the two-stream architecture, we use the RCAN [42] (stacking two RGs, and five RCABs in each RG) as our base model, and then we evaluate the impact of deploying the residual structure and adding a PPI edge infusion subbranch to our proposed PPIE-SSARN model in Table VII. Note that we replace the DDMF module with a

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TABLE VI
QUANTITATIVE COMPARISON WITH DIFFERENT DEMOSAICING METHODS IN TERMS OF PSNR, SSIM, AND SAM ON THE CHIKUSEI DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN **BOLD**

	WB	PPID	3D-ResNet	DPDNet	MCAN	Ours
PSNR↑	27.472	25.938	31.566	34.123	34.488	**36.947**
SSIM↑	0.783	0.684	0.906	0.941	0.944	**0.967**
SAM↓	26.184	31.451	16.026	12.026	11.477	**8.698**

TABLE VII
ABLATION STUDIES OF DIFFERENT NETWORK ARCHITECTURES

Residual Structure	PPI Edge Infusion	PSNR	SSIM	SAM
✗	✗	45.713	0.988	5.165
✓	✗	46.361	0.987	4.862
✓	✓	**47.263**	**0.991**	**4.142**

5×5 convolution layer to extract the initial features, ensuring that the effects are improved only due to the deployment of the two-stream architecture.

As shown in Table VII, the introduction of residual structure and PPI edge infusion subbranch is valuable in improving the quality of demosaiced multispectral images, leading to over 1.5 dB and 1.0 improvements in PSNR and SAM, respectively. It is proven that our proposed spatial–spectral residual learning structure can reduce the training difficulty and effectively improve the demosaicing ability since the traditional WB-based method offers reliable prior knowledge for the network.

2) Initial Feature Extraction Module: Effective extraction of preliminary features from the raw image has been proven to be an important processing step toward high-quality MSFA demosaicing [40]. To demonstrate the effects of our proposed DDMF module, we evaluate the performances of five initial feature extraction schemes as follows.

1) *Conv* 5×5*:* Using a standard 5×5 convolution layer with a stride of 1 to increase the dimension of the raw mosaic image to 16 channels.
2) *Sparse Split:* Using the split sparse image cube as the initial input of the network based on the design of [36].
3) *Sparse Split and Interpolation:* Interpolating the split sparse image as the initial network input based on the design of 3D-ResNet [39].
4) *MCM:* Using the MCM to extract preliminary features based on the design of MCAN [40].
5) *DDMF:* Using our proposed DDMF module to extract spatial and spectral-adaptive mosaic initial features.

For a fair comparison, all models use the same two-branch network architecture, trained for 4000 epochs with the same learning rate. As shown in Table VIII, using the split sparse image cube as the initial features will seriously undermine the accuracy for reconstructing the demosaiced images, demonstrating that sparse information is not conducive to recovering important details for mosaic images. For instance, the PSNR index significantly drops from 48.461 to 43.134 dB, while the SAM value adversely increases from 3.175 to 5.845. It is experimentally observed that the specially designed mosaic initial feature extraction methods, such as the MCM and our proposed DDMF, generally perform better. In comparison, our spatial- and spectral-adaptive weight-sharing mosaic preliminary feature extraction strategy can effectively avoid artifacts and aliasing problems and further improve the demosaic ability of the model.

TABLE VIII
ABLATION STUDIES OF DIFFERENT INITIAL FEATURE EXTRACTION SCHEMES

Initial Feature Extraction Strategy	PSNR	SSIM	SAM
Conv 5×5	47.263	0.991	4.142
Sparse Split	43.134	0.983	5.845
Sparse Split and Interpolation	46.098	0.989	4.992
MCM	47.521	0.992	3.717
DDMF	**48.461**	**0.994**	**3.175**

TABLE IX
ABLATION STUDIES OF DIFFERENT PPI INFORMATION EXTRACTION/INFUSION STRATEGIES

PPI Infusion Strategy	PSNR	SSIM	SAM
PPI (Repeat $\Rightarrow$ 16 channels)	47.551	0.992	3.834
PPI with Conv 3×3 (1 channel $\Rightarrow$ 16 channels)	47.648	0.991	3.743
PPI with Conv 3×3 (Repeat $\Rightarrow$ 16 channels)	47.763	0.992	3.579
PPI with Sobel Filter (Repeat $\Rightarrow$ 16 channels)	**48.461**	**0.994**	**3.175**

3) PPI Edge Infusion Strategy: Since the high-frequency information in PPI and demosaiced multispectral image exhibits a strong positive linear relationship [41], we adopt the Sobel filters to perform gradient extraction on the PPI as the additional compensation of edge-related information. To demonstrate the effects of our proposed PPI edge infusion strategy, we use the same two-branch network architecture as the base model and then investigate the demosaicing ability using different PPI information extraction/infusion strategies as follows.

1) *PPI (Repeat $\Rightarrow$ 16 Channels):* Repeating the PPI along spectral dimension to 16 channels as the infusion information.
2) *PPI With Conv* 3×3 *(One Channel $\Rightarrow$ 16 Channels):* Using a 3×3 convolution layer to increase the dimension of the PPI to 16 channels as the infusion information.
3) *PPI With Conv* 3×3 *(Repeat $\Rightarrow$ 16 Channels):* Using a 3×3 convolution layer to extract the information of the PPI and repeat the feature map along spectral dimension to 16 channels as the infusion information.
4) *PPI With Sobel Filter (Repeat $\Rightarrow$ 16 Channels):* Using the Sobel filters to calculate the gradient map of the PPI and repeat it along spectral dimension to 16 channels as our proposed PPI edge infusion strategy.

In Table IX, we prove that extracting and utilizing the edge-related information of PPI is helpful for the MSFA demosaicing task, since the high-frequency information in PPI and the reconstructed image present a strong linear relationship. In contrast, directly using the PPI or using the

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TABLE X
EFFICIENCY AND PARAMETER COMPARISON WITH DIFFERENT DEMOSAICING METHODS

Method	Params(M)	GFLOPs(G)	Running Time(ms)	
			CPU	GPU
WB	-	-	273.21	-
PPID	-	-	2227.43	-
3D-ResNet	0.45	1760.57	-	6.41
DPDNet	0.50	121.57	-	1.50
MCAN	1.37	71.76	-	1.19
Ours	0.92	223.73	-	2.14

convolution layer to extract additional information adaptively cannot achieve satisfactory results.

D. Efficiency and Parameter Comparison

We also provide computational cost and running time comparisons of different demosaicing methods, which is of great importance when applied in realistic multispectral imaging systems. We compare the CPU running time of two traditional approaches and model parameters, GFLOPs, and GPU running time of several state-of-the-art deep-learning-based approaches in Table X. All experiments are implemented using the same machine (Intel Xeon CPU E5-2680 v4 2.40 GHz, and NVIDIA GPU RTX 2080Ti) and carried out on 50 raw mosaic images with 480×512 spatial resolution. The running time is calculated by the average of the 50 testing images. As shown in Table X, our proposed PPIE-SSARN model can ensure a real-time demosaicing performance with a frame rate of over 460 frames/s.

V. CONCLUSION

In this article, we present a two-branch CNN-based network for MSFA image demosaicing. To achieve high-accuracy demosaicing of MSFA images, we design the PPIE-SSARN model, deploying a spatial–spectral adaptive residual compensation subbranch to help reconstruct demosaiced images. The PPI edge infusion subbranch enriches edge-related information by offering prior knowledge. Moreover, we propose an effective mosaic initial feature extraction module with a spatial- and spectral-adaptive weight-sharing strategy that can avoid artifacts and aliasing problems during the MSFA demosaicing process. Compared with the state-of-the-art conventional and deep learning-based MSFA demosaicing solutions, our proposed method generates more accurate demosaicing results in terms of spatial accuracy and spectral fidelity. A series of experiments demonstrate that our proposed PPIE-SSARN model outperforms existing MSFA demosaicing approaches in quantitative metrics and visual comparisons. The proposed method could be potentially utilized to build high-quality MSFA-based image acquisition systems working well in medical imaging, food quality inspection, and remote-sensing applications.

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